Practice Questions

Question 1

"Surely computers cannot be intelligent – they can do only what their programmers tell them". Is the latter statement true, and does it imply the former?

Question 2

To what extent are the following computer systems instances of artificial intelligence:

(i) Supermarket bar code scanners.
(ii) Web search engines
(iii) Voice-activated telephone menus
(iv) Internet routing algorithm that responds dynamically to the state of the of the network

Question 3

The **farmer's** problem is usually stated as follows: A farmer wants to cross the river and take with him a lion, a goat, and a yam. There is a boat that can fit himself plus either the lion, the goat, or the yam. If the lion and the goat are alone on one shore, the lion will eat the goat. If the goat and the yam are alone on the shore, the goat will eat the yam. How can the farmer bring the wolf, the goat, and the cabbage across the river?

a. Formulate the problem precisely, making only those distinctions necessary to ensure a valid solution. Draw diagram of the complete state space.

b. Implement and solve the problem optimally using an appropriate search algorithm. Is it a good idea to check for repeated states?

Question 4

Define the following terms:

i. Agent
ii. Agent Function
iii. Rationality
iv. Goal-based agent

Question 5

For each of the following activities, give a PEAS description of the task environment:

v. Playing football
vi. Shopping for used AI books on the internet
vii. Playing a tennis match
viii. Bidding on an item at an auction

Question 6

What is heuristic search? Describe how it is different from basic search.

Question 7

Define the following terms:

i. State
ii. State space
iii. Search tree
iv. Search node
v. Goal
vi. Action

Question 8

For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate.

i. An Agent that senses only partial information about the state cannot be perfectly rational.
ii. There exists a task environment in which every agent is rational.
iii. There exist task environments in which no pure reflex agent can behave rationally.
iv. The input to an agent program is the same as the input to the agent function.
v. Every agent function is implementable by some program/machine combination.
vi. Suppose an agent selects its action uniformly at random from the set of possible actions; there exists a deterministic task environment in which this agent is rational.
vii. It is possible for a given agent to be perfectly rational in two distinct task environments.
viii. Every agent is rational in an observable environment.

Question 9

i. Can there be more than one agent program that implements a given agent function? Give example, or show why one is not possible.
ii. Are there agent functions that cannot be implemented by any agent program? Explain.
iii. Give a fixed architecture, does each agent program implement exactly one agent function?
iv. Support we keep the agent program fixed but speed up the machine by a factor of two. Does that change the agent function?

Question 10

Which of the following propositional logic constructs are correct?

a. $\text{False} \vDash \text{True}$
b. $\text{True} \vDash \text{False}$

c. $(A \wedge B) \vDash (A \Leftrightarrow B)$
d. $A \Leftrightarrow B \vDash A \vee B$
e. $A \Leftrightarrow B \vDash \neg A \vee B$
f. $(A \wedge B) \Rightarrow C \vDash (A \Rightarrow C) \vee (B \Rightarrow C)$
g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$
h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \vDash (A \vee B)$
i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \vDash (A \vee B) \wedge (\neg D \vee E)$

Question 11

Consider the following facts:

1. The members of Agip Estate Friendship Club are Kalu, Ejiro, Sotonye, and Uju
2. Kalu is married to Ejiro
3. Sotonye is Uju's brother
4. The spouse of every married person in the club is also in the club
5. The last meeting of the club was at Kalu's house

a. Represent these facts in predicate logic
b. From the facts given above most people would be able to decide on the truth of the following additional statements:
   6. The last meeting of the club was at Ejiro's house
   7. Uju is not married
c. Can you construct a truth table proof to demonstrate the truth 6 and 7 given the 1-5? Do so if possible. Otherwise, add the facts you need and then construct the proofs.